Department of Mathematics
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Ablanvi Songo

Research Interests

My research focuses on nonlinear analysis, nonsmooth analysis, set-valued analysis, critical point theory, with applications to partial differential equations (PDEs), coupled nonlinear elliptic systems, differential inclusions, and coupled semilinear elliptic systems of differential inclusions using variational and topological methods. My research also explores applications of set-valued analysis to (N)-body problems in quantum physiscs. In addition, I am interested in exploring the potential application of applying my theoretical results and analytical tools to biological models formulated as sytems of differential equations or differential inclusions, as well as investigating how these analytics approaches can be combined with numerical methods for ODEs and PDEs.

Education

2021 -2026 **Ph.D. in Mathematics**, *Université de Sherbrooke, Canada*,
Title : Théorèmes de points critiques pour des fonctionnelles éventuellement non différentiables,
Supervisors: Fabrice Colin, Vasilisa Shramchenko
Grade: "Excellent"

2019-2020 **M.Sc. (1st year) in Actuarial and Financial Mathematics**, *Université de Rouen Normandie, France*

2018-2019 **M.Sc. (2nd year) in Fundamental and Applied Mathematics**, *Université de Rouen Normandie, France*

2017-2018 **M.Sc. (1st year) in Fundamental Mathematics**, *Aix Marseille Université, France*

2011-2015 **B.Sc. in Mathematics**, *Université de Lomé, Togo*

Publications and Preprints

Published journal articles

2026 **An infinite-dimensional Linking theorem with application to a system of coupled Poisson equations**, *with F. Colin, Journal of Mathematical Analysis and Applications, (2026), https://doi.org/10.1016/j.jmaa.2026.131036*

2026 **Existence result for a system of two semilinear coupled Poisson equations with asymptotically linear nonlinearities**, *Differential Equations & Applications, in Press*

2026 **An infinite dimensional Saddle Point Theorem and application**, *with F. Colin, Boundary Value Problems, (2026), https://doi.org/10.1186/s13661-026-02234-8*

Submitted journal articles

2026 **An infinite-dimensional Brouwer's fixed-point theorem**

2026 **An infinite-dimensional mountain pass theorem with applications to nonlinear elliptic systems**, *with F. Colin, https://doi.org/10.48550/arXiv.2512.02229*

2026 **On the Fountain Theorem for Continuous Functionals and Its Application to a Semilinear Elliptic Problem in R^2**, *with F. Colin, https://doi.org/10.48550/arXiv.2509.16059*

2026 **Linking theorems for multivalued functionals and application to partial differential inclusions**, *with F. Colin, https://doi.org/10.48550/arXiv.2607.22506*

Preprints

2026+ **Subcritical elliptic systems for partial differential inclusions in dimension two**, *in preparation*

Other Scientific archievements

2019 **Master's (M.Sc.) thesis**, *Université de Rouen Normandie, France*,
Title: Les équations d'Hamilton et Jacobi sur les réseaux : Sous et Sur-solutions à flux-limité,
Grade : "Très-bien"

Last updated: September 9, 2026

Talks

May 2026 **XXVIIIe colloque panquébécois de l'Institut des Sciences Mathématiques**, *UTQR*, Trois-Rivières

April 2026 **Présentations "180 secondes"**, *Caféfak, Faculté des sciences, Université de Sherbrooke*

November 2025 **Séminaire Junior en Géométrie et Physique Mathématique**, *Département de mathématiques, Université de Sherbooke*

May 2025 **XXVIIe colloque panquébécois de l'Institut des Sciences Mathématiques**, *McGill University*

February 2025 **Club Math**, *Department of Mathematics, Université de Sherbooke*

Nov 2023 **Séminaire "Séminaire" d'étudiants gradués**, *Department of Mathematics, Université de Sherbooke*

Awards and Grants

2021-2024 **Doctoral scholarship of the Faculty of Sciences**, *Faculté des sciences, Université de Sherbrooke*

2021-2026 **ISM Travel Grant**, *Institut des Sciences Mathématiques (ISM), Montréal*

Hiver 2026 **Award for Outstanding First Publication**, *Faculté des sciences, Université de Sherbrooke*

2023-2024, 2022-2023 **ISM Graduate Scholarship**, *Institut des Sciences Mathématiques (ISM), Montréal*

2022-2023 **Bourse In memorian Ernest-Monga for perseverance and commitment**, *Departement of Mathematics, Université de Sherbrooke*

2018-2019, 2017-2018 **Scholarship of Government of Togo**, *Ministère de l'Enseignement Supérieur et de la Recherche*

Conferences and Schools Attended

May 2026 **XXVIIIe colloque panquébécois de l'Institut des Sciences Mathématiques**, *UTQR*, Trois-Rivières

November 2025 **Séminaire Junior en Géométrie et Physique Mathématique**, *Département de mathématiques, Université de Sherbooke*

October 2025 **3rd University of North Carolina at Greensboro Virtual PDE Conference : dedication to Professor Alfonso Castro in celebration of his 75th birthday and his outstanding contributions to Differential Equations**, *Departement of Mathematics and Statistics (online)*

May 2025 **XXVIIe colloque panquébécois de l'Institut des Sciences Mathématiques**, *McGill University*

June 2024 **XXVIe colloque panquébécois de l'Institut des Sciences Mathématiques**, *Université de Québec à Montréal*

October 2023 **Black Heroes of Mathematics Conference 2023**, *ICMS, the University of Edinburgh (online)*

June 2023 **XXVe colloque panquébécois de l'Institut des Sciences Mathématiques**, *Université de Sherbrooke*

July 2022 **Topological Methods in Nonlinear Analysis: Recent Advances - Conference in memory of Professor Andrzej Granas**, *Centre de recherches mathématiques (CRM), Université de Montréal*

May 2022 **XXIVe colloque panquébécois de l'Institut des Sciences Mathématiques**, *Université Laval*

Teaching Experience

Université de Sherbrooke

Winter 2026 **Teaching Assistant**, *MAT 117 (Differential and Integral Calculus of Several Variables and Vector Analysis)*

Fall 2025 **Teaching Assistant**, *MAT 098 (Supplementary Material on Integral Calculus)*

Summer 2025 **Teaching Assistant**, *MAT 455 (Analysis III - Differential and Integral Calculus in $\mathbb{R}^n$)*

Winter 2025 **Teaching Assistant**, *MAT 129 (Analysis I - Elements of Analysis)*

Fall 2024 **Teaching Assistant**, *MAT 199 (Linear Algebra Applied to Computer Science), MAT 221, CQP 208*

Winter 2024 **Teaching Assistant**, *MAT 129, MAT 221 (Differential and Integral Calculus), CQP 208*

Fall 2023 **Teaching Assistant**, *MAT 221 (Differential and Integral Calculus)*

Winter 2023 **Teaching Assistant**, *MAT 129 (Elements of Analysis)*

Fall 2022 **Teaching Assistant**, *MAT 221 (Differential and Integral Calculus), MAT 324*

Summer 2022 **Teaching Assistant**, *MAT 455 (Analysis III)*

Winter 2022 **Teaching Assistant**, *MAT 193 (Linear Algebra)*

Fall 2021 **Teaching Assistant**, *MAT 324 (Mathematical Models)*

Winter 2021 **Teaching Assistant**, *MAT 193 (Linear Algebra)*

Lycée de Davié (Togo)

2014-2017 **Nationally certified teacher**, *Mathematics, Physics-Chemistry*

Academic and Community Involvement

March 2026 **Volunteer during the presentation of mathematical games at the Sherbrooke Museum of Nature and Science organized by our Department**, *Sherbrooke*

May 2025 **Volunteer during the final round of the International Mathematical and Logical Games Championship (province of Quebec)**, *Cegep de Sherbrooke*

Mars 2025 **Volunteer during the semi-final round of the International Mathematical and Logical Games Championship (Sherbrooke Region)**, *Department of Mathematics, Université de Sherbrooke*

March 2025 **Volunteer during the presentation of mathematical games at the Sherbrooke Museum of Nature and Science organized by our Department**, *Sherbrooke*

March 2024 **Volunteer during the semi-final round of the International Mathematical and Logical Games Championship (Sherbrooke Region)**, *Department of Mathematics, Université de Sherbrooke*

March 2024 **Volunteer during the presentation of mathematical games at the Sherbrooke Museum of Nature and Science organized by our Departement**, *Sherbrooke*

May 2023 **Volunteer during the Quebec final of the International Mathematical and Logical Games Championship**, *Department of Mathematics, Université de Sherbrooke*

Winter 2023 **Volunteer during the Faculty of Science Open House**, *Université de Sherbrooke*

Fall 2022 **Volunteer during the Faculty of Science Open House**, *Université de Sherbrooke*

2021–2023 **Volunteer mathematics tutor for Le Diplôme Avant la Médaille (DAM) academic support program for student-athletes**

Technical Strengths and Skills

Programming

Python, SQL, LaTeX

Languages

French, English, Ewè

Academic References

Fabrice Colin, *Associate Professor*, Laurentian University
Email: fcolin@laurentian.ca

Vasilisa Shramchenko, *Full Professor*, Université de Sherbrooke
Email: Vasilisa.Shramchenko@USherbrooke.ca